



BLUE CHRISTMAS

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SEASONAL AFFECTIVE DISORDER (SAD)

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Depression developing in autumn/winter which remits in spring/summer and which is not a reaction to a regular winter stressor, eg seasonal unemployment!

Plus

- Increased duration of sleep
- Increased appetite
- Carbohydrate craving
- Weight gain



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In SAD the physiological symptoms of depression are the reverse of classic depression, i.e. increased sleep, increased appetite, increased weight and mood not worse in the mornings.

But cognitive and emotional symptoms are identical

DSM-IV CRITERIA FOR MAJOR DEPRESSIVE DISORDER

1) The symptoms cause clinically significant distress or impairment in social or occupational functioning

2) Low mood or loss of interest or pleasure for most of the day for two weeks plus at least 3 of the following

- Change of weight
- Sleep disturbance nightly
- Agitation or retardation daily
- Fatigue / lack of energy daily
- Guilt / worthlessness
- Poor concentration and indecisiveness
- Recurrent thoughts of death or suicide

DSM-IV CRITERIA FOR SEASONAL AFFECTIVE DISORDER

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SAD is a mood disorder coded with a Seasonal Pattern Specifier:

- A. regular temporal relationship between onset of major depressive episodes and time of year**
- B. full remissions at a specific time of year**
- c. two episodes in past two years without non-seasonal episodes**
- D. seasonal episodes exceed non-seasonal episodes over the lifetime**

Tendency for mood to improve with...

- Travel to lower latitudes
- Living/working in higher illumination



In Contrast To Non-seasonal Depression There Is Less...

- Suicidal thoughts
- Morning worsening of mood

How common is SAD?

SPAQ - Seasonal Pattern Assessment Questionnaire

SPAQ prevalence 1-10 % of population of North America

Using DSM-IV criteria 1-3 %

SPAQ overestimates prevalence of SAD because of poor reliability of

- Recall of depressive episodes
- Recall of seasonal patterns

SAD is:

- Commoner in women
- Commoner in younger ages
- Commoner in North America than Europe



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Low mood is on a continuum ranging from

Unhappiness (transient/permanent)  mild depression
 major depression  depressive psychosis
in SAD low mood is generally mild-moderate

In severe depression up to 13% commit suicide
(20 x more likely than the general population)

in SAD completed suicide is rare

In SAD depressive psychotic symptoms are very rare

Summer

- ± Mild Hypomania
- ± Relapse if weather poor or reduced ambient light exposure

Family History

1st degree relatives often have:

- **Mood disorders**
- **Alcoholism**
- **SAD**

Impairment of function

Work, especially safety-critical work, and relationships affected by daytime tiredness and fatigue

Course of SAD

- Age of onset: 20-30
- After about 10 years about 40% progress to non-seasonal depression
- 40% continue to have SAD
- Up to 20% remit completely

The best predictor of season-bound recurrence is presence of atypical depressive symptoms, ie

- increased sleep
- increased appetite
- increased weight

Latitude

- SAD is due to lack of sunlight in winter

But

- No clear latitude gradient – other factors have to be considered:
 - ❖ Hours of sunshine in winter
 - ❖ Out of door vs indoor employment
 - ❖ Day-length

Certain ethnic groups who live in high latitudes, eg Lapps and Icelanders, may have adapted to the long arctic winter

What Is Special About SAD?

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Seasonality - SAD is at the extreme end of the spectrum of seasonality

Atypical neurovegetative depressive symptoms -

Increased sleep

Increased appetite

Carbohydrate craving

Increased weight

Resembles winter behaviour of many mammals, to achieve maximum energy conservation

SAD also resembles bulimia nervosa

Responsiveness to light treatment -



Can-SAD STUDY (Lam, 2006)

- RCT of light therapy vs fluoxetine
- 8 weeks of
 - 1) 10,000 lux 30 mins in morning + placebo capsuleor
 - 2) 100 lux (placebo) + fluoxetine 20 mg daily



VS



Can-SAD STUDY (Lam, 2006) cont'd

96 patients

- 50 % of light treatment patients remitted
- 54 % of fluoxetine patients remitted

No significant difference

Advantages of Light Treatment

- Earlier response
- Lack of treatment-emergent adverse events, ie agitation, sleep disturbance, palpitations.

Reference

Lam et al, American Journal of Psychiatry, 2006, 163: 805-812

PATHOPHYSIOLOGY OF SAD

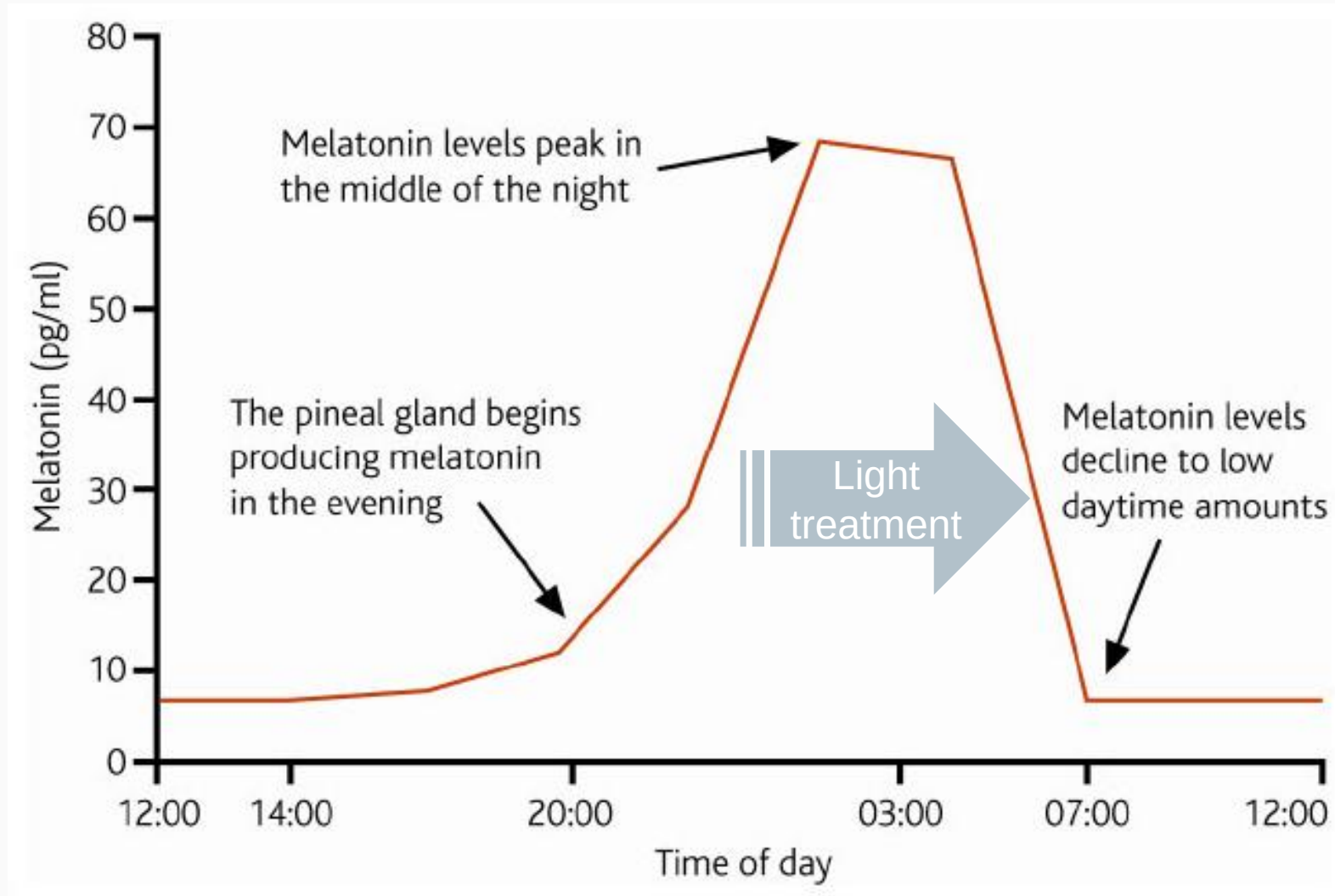
(Sohn and Lam, 2005)

- Circadian rhythms
- Neurotransmitters
- Molecular genetics
- Response to bright light

HYPOTHESIS

Seasonal depression is attributed to circadian rhythm abnormalities, i.e. when dark / light cycle is out of synch with sleep / wake cycle.

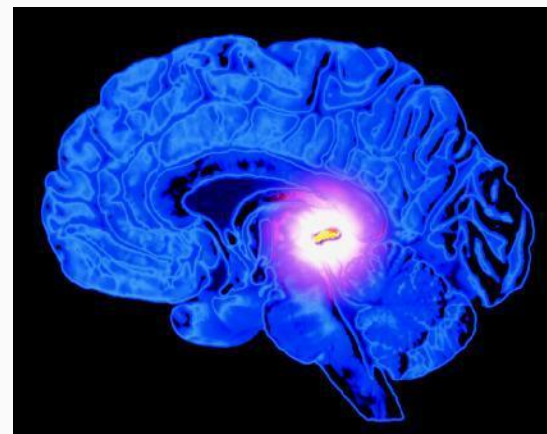
Light therapy resets the body clock by moving back the hands of the clock to where they should be to match light / dark and awake / asleep cycles.



PINEAL

Secretes melatonin under the influence of

- Suprachiasmatic Nucleus (S.C.N.) – a circadian “clock” effect
- External light exposure which suppresses melatonin release directly

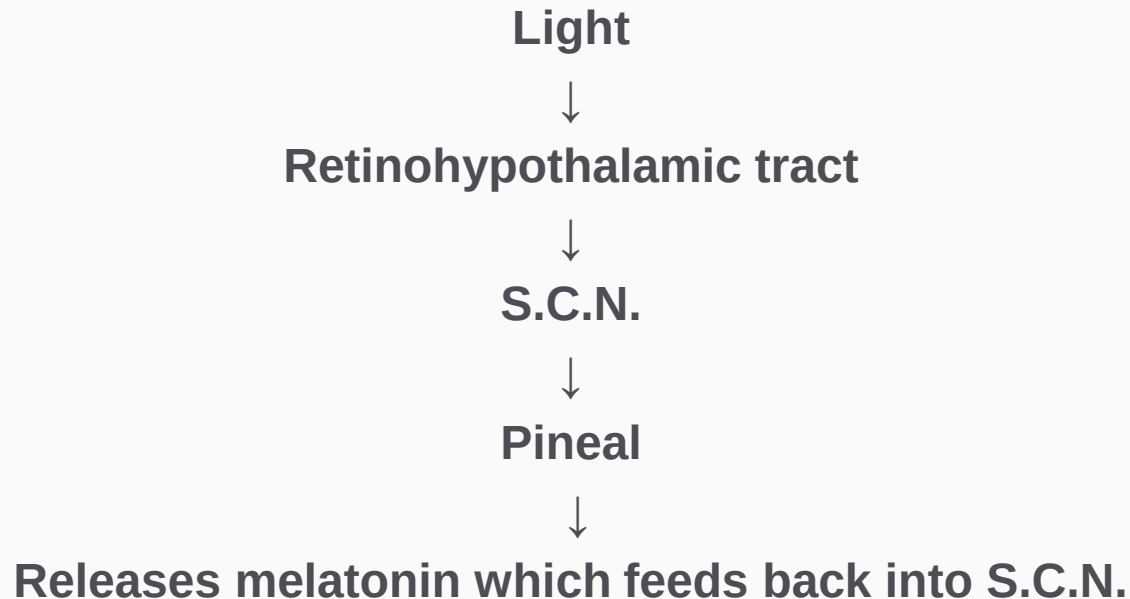


Melatonin is secreted at high levels at night and low levels in daytime.

Melatonin levels begin to rise at 8 pm, ie when light is diminishing.

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The central pacemaker that synchronises internal circadian rhythms with external time cues, e.g. light is in the S.C.N.



Seasonal Affective Disorder (SAD) - cont'd

LIGHT SHIFTS CIRCADIAN RHYTHMS

- Bright light in mornings ADVANCES the release phase of melatonin so that melatonin release occurs earlier
- Shifting the phase of one circadian rhythm, ie melatonin release, changes the time interval to another circadian rhythm (eg, sleep-wake cycle). This is the phase-shift hypothesis.



- **Common**
- **There is a spectrum ranging from "winter blues" to SAD with "atypical" biological symptoms**
- **Depression is mild / moderate in SAD**
- **Relatively low suicide risk in SAD**
- **Differential diagnosis:**
non-seasonal depression * CFS I "ME" * Bulimia nervosa * hypothyroidism
- **SAD can cause occupational disability in winter months especially for safety-critical work**
- **SAD is treatable with phototherapy or antidepressants**
- **Occupational Health departments could provide the apparatus with instructions**

“You are planning for winter pressures in your emergency department. You wonder if you will need additional psychiatric support over the Christmas period to cope with a perceived increase in the number of suicide attempts. When you bring this up at a senior team meeting a colleague suggests that in fact the number of suicides decreases over the Christmas period. You wonder if this is true”.

EDINBURGH (Masterton et al, 1991)

All Parasuicidal Admissions - January 1969 to December 1987

- Over 22 Thousand
- 20 % decrease for 4th December – 1st January
- 11 % increase for first week of New Year

WHO 13 EUROPEAN CENTRES (Ajdacic-Gross et al, 2003)

Suicide Attempts 1989-1996

- Over 24 Thousand
- Week before Christmas - statistically fewer attempts
- Christmas Day – average
- December 27th 39 % increase

SWITZERLAND

Completed Suicides 1969-1994

- Over 37 thousand
- Suicide rates in December 10 % less than average for the year
- December dates with lowest rates of suicide 23rd, 25th and 30th December



- Lower rates of suicide / deliberate self-harm over Christmas period
- Higher rates after Christmas
- ? “ Postponement effect” ?

